

ANIL CHANNAPPA JIRAGYALE

Full Stack AI Engineer | LLM & RAG Systems | Backend & API Development | React.js / Next.js

+91 9591585862 • aniljiragyale07@gmail.com • Bangalore, India • [linkedin.com/in/anil-jiragyale](https://www.linkedin.com/in/anil-jiragyale) • github.com/aniljiragyale • portfolio-anil-two.vercel.app

PROFESSIONAL SUMMARY

Full Stack AI Engineer with enterprise experience building AI-powered platforms, LLM-integrated workflows, vector database systems, and scalable REST APIs at GSK GCC and Rokkun Systems. Delivered 3+ production platforms serving 100+ users — improving query accuracy by 30%, reducing manual effort by ~40%, and enabling large-scale semantic search using Milvus, LangChain, and LangGraph. Skilled in React.js, Next.js, Python (FastAPI), Node.js, PostgreSQL, AWS, Azure, RAG architecture, and prompt engineering.

TECHNICAL SKILLS

Languages:	Python, JavaScript (ES6+), Java, SQL
Frontend:	React.js, Next.js, Tailwind CSS, HTML5, CSS3, Responsive UI Design
Backend:	FastAPI, Node.js, Express.js, Spring Boot, REST APIs, Authentication, API Integration
AI / ML:	LLM Integration, LangChain, LangGraph, RAG Architecture, Prompt Engineering, Semantic Search, Vector Embeddings, NLP, Machine Learning, TensorFlow, Scikit-learn
Databases:	PostgreSQL, MySQL, MongoDB, DuckDB, Milvus (Vector DB)
Cloud & DevOps:	AWS, Azure App Service, Git, GitHub, Jenkins, CI/CD Pipelines, Linux, Postman
Data & BI:	Power BI, Pandas, NumPy, Matplotlib, Data Visualization

PROFESSIONAL EXPERIENCE

Full Stack AI Engineer

Apr 2026 – Present

Rokkun System Private Ltd | Bangalore, India

- Architected and enhanced an AI-powered market intelligence platform integrating news ingestion, semantic processing, and LLM-based retrieval pipelines using LangChain and LangGraph.
- Integrated Milvus vector database for large-scale semantic search, optimizing chunking strategies and embedding pipelines for high-accuracy, low-latency document retrieval.
- Developed a semantic search tool with time-decay ranking and natural language time filtering, improving retrieval accuracy across 10,000+ indexed documents.
- Built data processing workflows including scraping fallback handling, content cleaning, and optimized chunking for vector embeddings, reducing pipeline failure rate by ~35%.
- Designed and implemented a user feedback loop for AI-generated insights — covering ratings, voting, and comments — enabling continuous model evaluation and improvement.
- Built end-to-end feedback infrastructure with REST APIs (FastAPI/Node.js), scheduled jobs, and data pipelines feeding into model retraining and insight quality dashboards.
- Developed automated Actuals Comparison workflows to evaluate AI predictions against real-world outcomes, improving model transparency and stakeholder trust.
- Implemented Monthly Reporting pipelines summarizing system performance metrics and user feedback trends for product and engineering leadership.
- Enhanced frontend UX in Next.js — integrating feedback panels, fixing interaction issues, and aligning UI with product requirements.
- Improved system reliability by adding exception handling and fallback mechanisms using LLM-based research workflows; resolved production-level Milvus retrieval issues.
- Collaborated cross-functionally to define requirements, debug production issues, and deliver end-to-end features on schedule.

Graduate Trainee – AI & Full Stack Engineering

Aug 2024 – Mar 2026

GlaxoSmithKline (GSK) GCC | Bangalore, India

- Engineered 2 production enterprise platforms — CDM Accelerator and CDM Intake — delivering full-stack: React.js + Tailwind CSS frontend, FastAPI REST APIs, AI/LLM integrations, and Azure App Service deployments serving 100+ users.
- Implemented LLM-powered query processing with guardrail validation and RAG architecture, improving query resolution accuracy by 30% and reducing retrieval time by ~50%.
- Developed ASK CDF — an AI chatbot for natural language querying of clinical documentation — processing 200+ query types using RAG and prompt engineering.
- Built an AI-driven ER diagram generator converting unstructured clinical documents into structured data models using NLP and LLM pipelines, saving 6+ hours of manual effort per sprint.
- Designed a hybrid rule-based + AI validation engine for CSV/Excel data quality, reducing manual review effort by ~40% and cutting data error rates by 25%.
- Developed 10+ scalable FastAPI backend services for data ingestion, transformation, and analytics with secure authentication and REST API endpoints.
- Deployed applications on Azure App Service; managed CI/CD pipelines via Jenkins and GitHub, enabling zero-downtime deployments and reducing release cycle time by ~30%.
- Built Power BI dashboards and React-based reporting interfaces tracking 15+ KPIs, improving operational decision speed for 3+ business teams.
- Debugged and resolved 20+ production-level issues; collaborated in Agile sprints and code reviews with cross-functional teams.

Apprenticeship Trainee – Full Stack & DevOps

Oct 2023 – Jul 2024

GlaxoSmithKline (GSK) GCC | Bangalore, India

- Developed and maintained 5+ MERN stack (MongoDB, Express.js, React.js, Node.js) web application modules in a large-scale enterprise environment.
- Automated 3+ data processing workflows using Python scripting, reducing manual reporting effort by ~60% and saving 4+ hours per week.
- Conducted automation testing with 50+ test cases, achieving zero critical regression defects in final delivery.

KEY PROJECTS

Tariff Prediction Model – AI/ML System

Apr 2026 – Present

Rokkun System Private Ltd

- Developing an AI-powered tariff prediction model using time-series forecasting and machine learning to predict trade tariff outcomes from real-world economic and policy datasets.
- Designing data ingestion pipelines, feature engineering workflows, and model evaluation frameworks to benchmark prediction accuracy across multiple forecasting algorithms.
- Integrating the prediction model with the market intelligence platform to surface actionable tariff risk insights for end users.

CDM Accelerator – Enterprise AI Data Platform

2024 – 2026

GSK GCC

- Architected end-to-end AI-powered Clinical Data Management platform: ASK CDF (RAG chatbot), ER Extractor, and S2T Validator — deployed to 100+ enterprise users at GSK GCC.
- Built React.js + Tailwind CSS frontend, FastAPI backend APIs, and LLM services with guardrail-based hallucination control; improved query resolution accuracy by 30%.
- Deployed on Azure App Service with Jenkins CI/CD; integrated Power BI dashboards tracking 15+ KPIs, reducing clinical team decision latency by ~2 hours per sprint.

CDM Intake – Enterprise Workflow Automation

2024 – 2026

GSK GCC

- Built CDM Intake platform for structured clinical data intake, validation, and routing — React.js UI, FastAPI/Node.js REST APIs, MySQL/MongoDB — reducing intake processing time by ~35%.
- Designed AI + rule-based validation engine cutting manual review cycles by 40% and improving data governance compliance across 3 cross-functional teams.

Airline Reservation System – Full Stack Web App

Academic Project

- Built a full-stack airline reservation system using React.js, Java (Spring Boot), MySQL, and REST APIs — supporting flight search, seat booking, and ticket tracking for 500+ simulated bookings.

COVID-19 Spread Prediction – ML / Deep Learning

Academic Project

- Developed a predictive ML model achieving 85% accuracy on epidemiological datasets by benchmarking LASSO Regression, ARIMA, and LSTM models across 12 months of training data.

EDUCATION

B.E. Computer Science & Engineering | CGPA: 7.44 / 10.0

East Point College of Engineering and Technology | Bangalore, Karnataka

Feb 2020 – Jun 2024

Pre-University (PUC) – Science | Aggregate: 85%

Ekalavya Science PU College | Harugeri, Karnataka

Jun 2018 – Mar 2020

Secondary School (SSLC) | Aggregate: 70%

Vidyavardhak High School | Athani, Karnataka

2017 – 2018

CERTIFICATIONS & TRAINING

- AI with Machine Learning – Final Exam | Oracle Academy | Dec 2023
- Database Foundations | Oracle Academy | Nov 2023
- Java Foundations – Final Exam | Oracle Academy | Oct 2022
- Fundamentals of AI and ML | Infosys Springboard | Nov 2023
- Big Data 101 (BD0101EN) | IBM SkillsBuild | Dec 2023
- Certified Python Developer Associate (CPDA-24) | PDAC | Lifetime
- Step into RPA (Robotic Process Automation) | UiPath Academy | Nov 2023
- Java Engineer Hiring Challenge – Certificate of Excellence | TechGig / Persistent | Sep 2024